



Project Title: Electricity Bill System

Course: Python

Instructor: Dr. Salah Alghyaline

1. Team Members:

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JupyterLab Python 3 (ipykernel)

[2]: students = {
    "S001": {"name": "Ahmad Ali", "courses": ["CS101", "MATH201"]},
    "S002": {"name": "Sara Mahmoud", "courses": ["ENG150"]},
    "S003": {"name": "Omar Hussein", "courses": ["CS101", "ENG150", "PHYS110"]},
    "S004": {"name": "Lina Khaled", "courses": set()},
    "S005": {"name": "Yousef Nasser", "courses": ["CS101"]}
}

courses = {
    "CS101": {"name": "Introduction to Programming", "capacity": 3, "students": ["S001", "S003", "S005"]},
    "MATH201": {"name": "Calculus II", "capacity": 2, "students": ["S001"]},
    "ENG150": {"name": "Academic Writing", "capacity": 4, "students": ["S002", "S003"]},
    "PHYS110": {"name": "General Physics", "capacity": 2, "students": ["S003"]}
}

[3]: def add_student():
    student_id = input("Enter student ID: ")
    if student_id in students:
        print("Student ID already exists.")
    else:
        name = input("Enter student name: ")
        students[student_id] = {"name": name, "courses": set()}
        print("Student added successfully.")

def add_course():
    course_code = input("Enter course code: ")
    if course_code in courses:
        print("Course code already exists.")
    else:
        course_name = input("Enter course name: ")
        capacity = int(input("Enter course capacity: "))
        courses[course_code] = {"name": course_name, "capacity": capacity, "students": []}
        print("Course added successfully.")

def display_all_students():
    print("--- Students List ---")
    for s_id in students:
        print(f"ID: {s_id}, Name: {students[s_id]['name']}, Courses: {students[s_id]['courses']}")

def display_all_courses():
    print("--- Courses List ---")
    for c_code in courses:
        print(f"Code: {c_code}, Name: {courses[c_code]['name']}, Capacity: {courses[c_code]['capacity']}")
```

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JupyterLab Python 3 (ipykernel)

[2]: students = {
    "S001": {"name": "Ahmad Ali", "courses": ("CS101", "MATH201")},
    "S002": {"name": "Sara Mahmoud", "courses": ("ENG150")},
    "S003": {"name": "Omar Hussein", "courses": ("CS101", "ENG150", "PHYS110")},
    "S004": {"name": "Lina Khaled", "courses": set()},
    "S005": {"name": "Yousef Nasser", "courses": ("CS101")}
}

courses = {
    "CS101": {"name": "Introduction to Programming", "capacity": 3, "students": ["S001", "S003", "S005"]},
    "MATH201": {"name": "Calculus II", "capacity": 2, "students": ["S001"]},
    "ENG150": {"name": "Academic Writing", "capacity": 4, "students": ["S002", "S003"]},
    "PHYS110": {"name": "General Physics", "capacity": 2, "students": ["S003"]}
}

[3]: def add_student():
    student_id = input("Enter student ID: ")
    if student_id in students:
        print("Student ID already exists.")
    else:
        name = input("Enter student name: ")
        students[student_id] = {"name": name, "courses": set()}
        print("Student added successfully.")

    def add_course():
        course_code = input("Enter course code: ")
        if course_code in courses:
            print("Course code already exists.")
        else:
            course_name = input("Enter course name: ")
            capacity = int(input("Enter course capacity: "))
            courses[course_code] = {"name": course_name, "capacity": capacity, "students": []}
            print("Course added successfully.")

    def display_all_students():
        print("--- Students List ---")
        for s_id in students:
            print("ID:", s_id, "Name:", students[s_id]["name"], "Courses:", students[s_id]["courses"])

    def display_all_courses():
        print("--- Courses List ---")
        for c_code in courses:
            print("Code:", c_code, "Name:", courses[c_code]["name"], "Capacity:", courses[c_code]["capacity"])
```

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JupyterLab Python 3 (ipykernel)

def display_all_students():
    print("--- Students List ---")
    for s_id in students:
        print("ID:", s_id, "Name:", students[s_id]["name"], "Courses:", students[s_id]["courses"])

def display_all_courses():
    print("--- Courses List ---")
    for c_code in courses:
        print("Code:", c_code, "Name:", courses[c_code]["name"], "Capacity:", courses[c_code]["capacity"])

[4]: def main():
    while True:
        print("\n--- University Registration System ---")
        print("1. Add Student")
        print("2. Add Course")
        print("3. Register Course (Stage 2)")
        print("4. Drop Course (Stage 2)")
        print("5. Show Student Courses")
        print("6. Show Course Students")
        print("7. Display All Students")
        print("8. Display All Courses")
        print("9. Exit")

        choice = input("Enter your choice: ")

        if choice == "1":
            add_student()
        elif choice == "2":
            add_course()
        elif choice == "7":
            display_all_students()
        elif choice == "8":
            display_all_courses()
        elif choice == "9":
            print("Exiting...")
            break
        else:
            print("Invalid choice or feature not implemented yet.")

    # Run the program
    main()

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course (Stage 2)
4. Drop Course (Stage 2)
```

main()

```
--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course (Stage 2)
4. Drop Course (Stage 2)
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice: 7
--- Students List ---
ID: S001 Name: Ahmad Ali Courses: {'CS101', 'MATH201'}
ID: S002 Name: Sara Mahmoud Courses: {'ENG150'}
ID: S003 Name: Omar Hussein Courses: {'CS101', 'ENG150', 'PHYS110'}
ID: S004 Name: Lina Khaled Courses: set()
ID: S005 Name: Youssef Nasser Courses: {'CS101'}

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course (Stage 2)
4. Drop Course (Stage 2)
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice: 1
Enter student ID: 3230601
Enter student name: ahmad sami
Student added successfully.

--- University Registration System ---
1. Add Student
2. Add Course
3. Register Course (Stage 2)
4. Drop Course (Stage 2)
5. Show Student Courses
6. Show Course Students
7. Display All Students
8. Display All Courses
9. Exit
Enter your choice: 9
Exiting...
```

[]: